

Thesis Project
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Characterization of Long Bent Silicon Crystals for Channelling Applications at the LHC

Update 24 August 2026

WHAT IS 'CHANNELING' IN CRYSTALS?

THE PHYSICS

- *Channeling effect*

Charged particles get trapped and guided between regular atomic planes

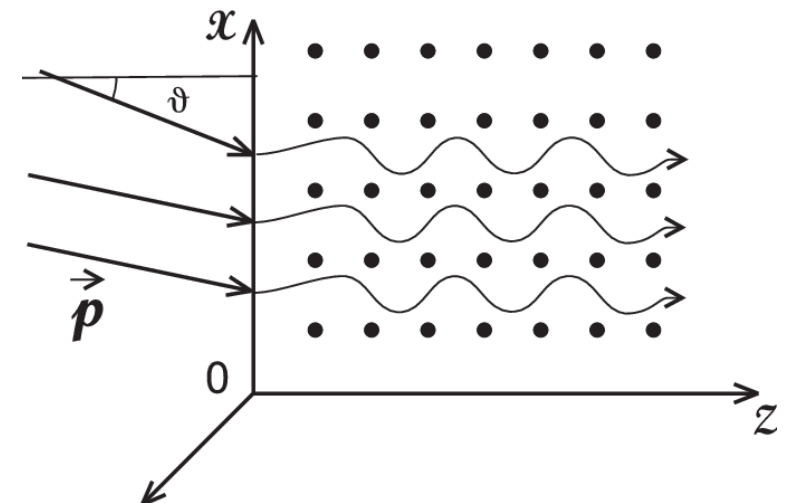
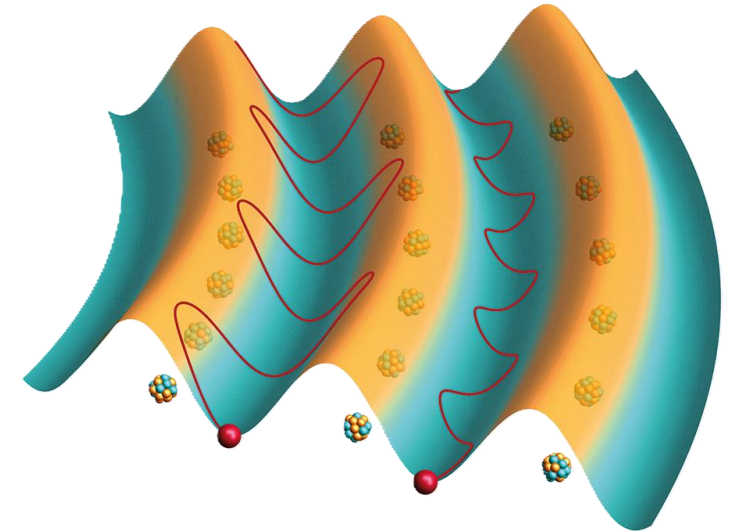
- *Beam deflection*

Bending the crystal physically steers the beam's trajectory

- *Advantage*

A compact alternative to traditional magnets for deflecting or extracting beams

Entry condition: the Lindhard angle θ_L
a particles enters the channel only if its incidence angle is
within $\pm 1/2 \theta_L$ of the crystal plane
 $\approx 13 \mu\text{rad}$ @ 180 GeV



THE TWOCRYST EXPERIMENT

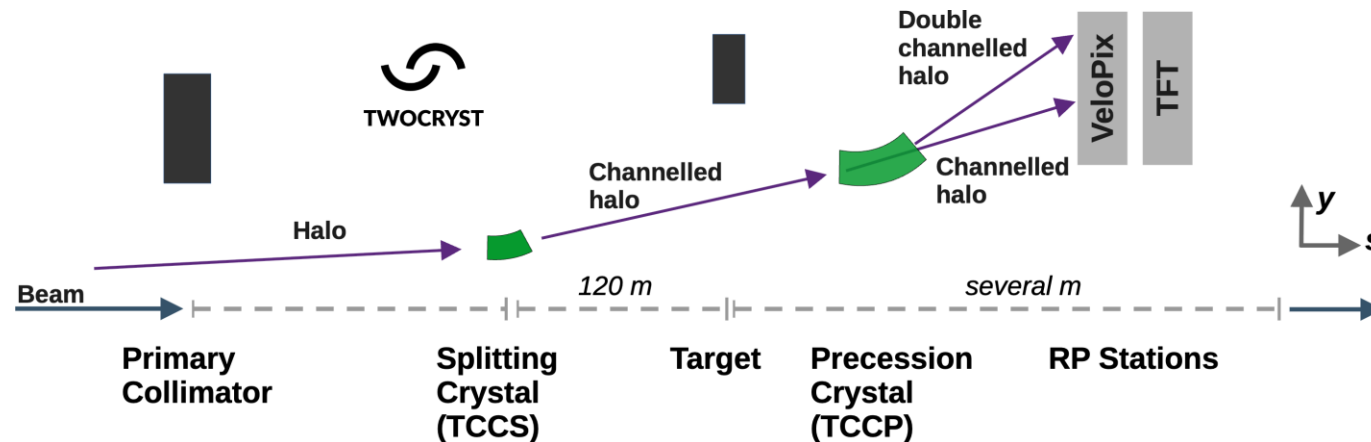
THE PROJECT

before

Bent crystals only used in collimation systems to deflect particles in the beam halo

(2025 - now)

Validate the **operational feasibility, crystal performance**, required for a double-crystal fixed-target experiment inside the LHC (*TeV scales*)



future developments

ALADDIN: exploits the strong electromagnetic fields inside bent silicon crystals to steer the Λ_c^+ particles to induce a strong deflection (spin precession)

①

DATA

2024 dataset for two long crystals

Taken at SPS-H8 test bench with 180 GeV beam (70% protons and 30% pions)

②

GOAL OF MY WORK

Measure the channeling efficiency ϵ_{ch} and the crystal torsion on newly fabricated long crystals.

③

WHAT IS NEW

These crystals were fabricated directly by CERN, unlike the ones mounted on TWOCRIST.

MY ANALYSIS

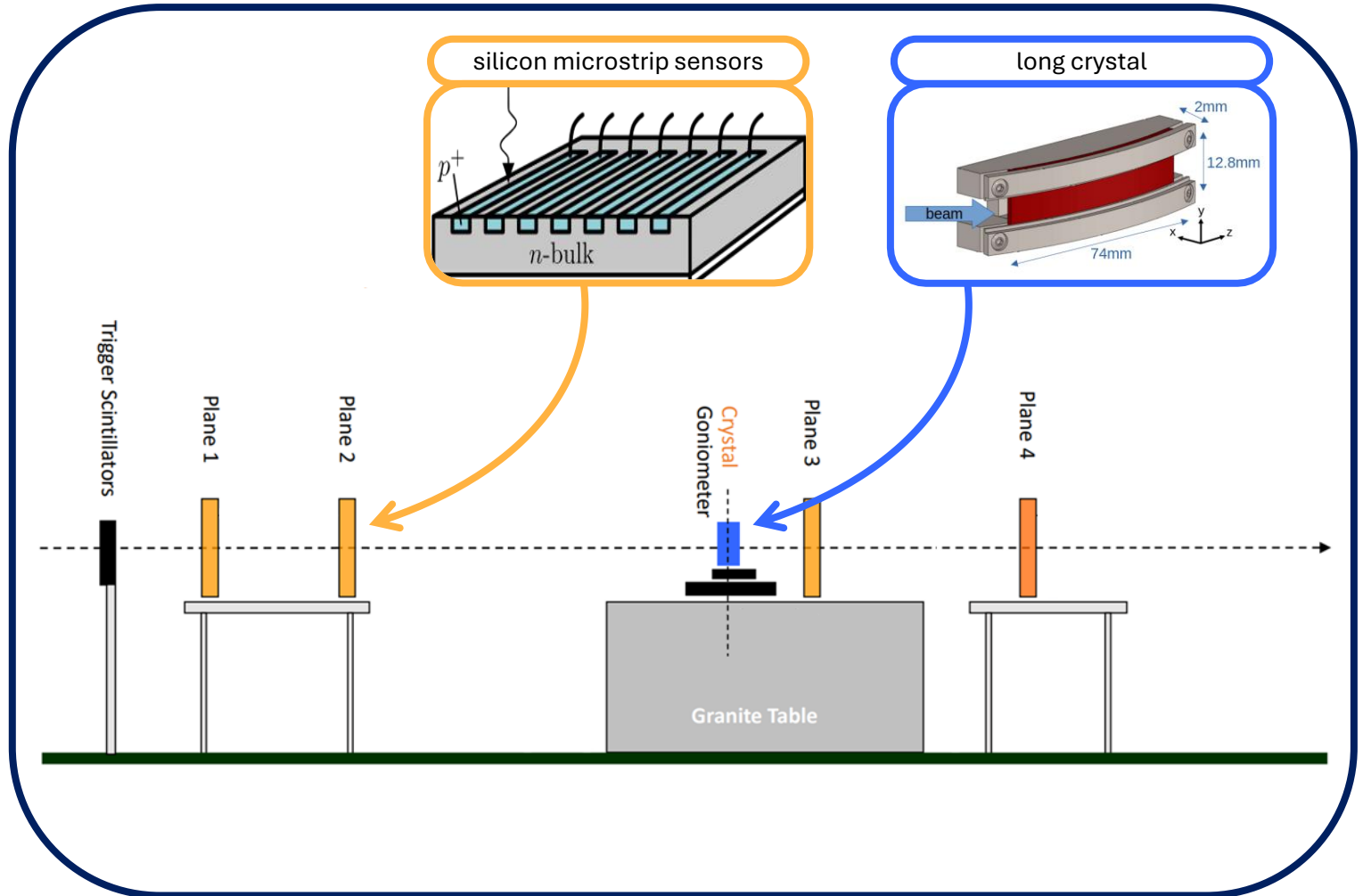
EXPERIMENTAL SETUP

1

DATA

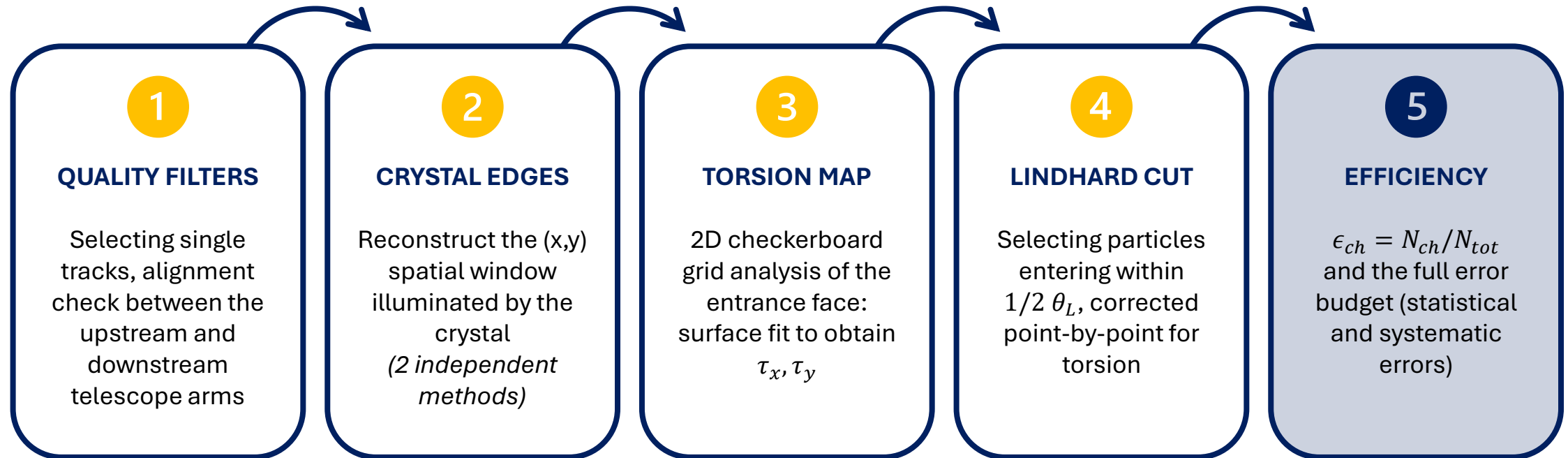
2024 dataset for two long crystals

Taken at SPS-H8 test bench with 180 GeV beam (70% protons and 30% pions)



THE ANALYSIS PIPELINE

METHOD



Analysis implemented in Python on RDataFrame (ROOT).

ANALYSIS: DETECTOR ALIGNMENT

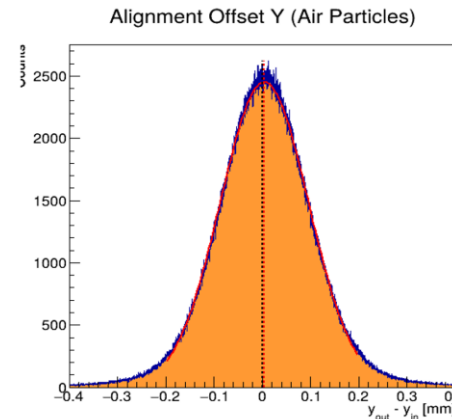
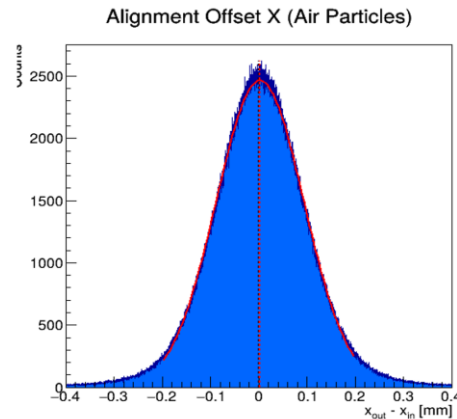
PRELIMINARY ANALYSIS

1

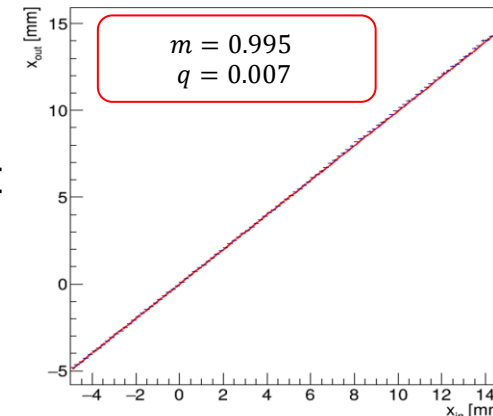
QUALITY FILTERS

Selecting single tracks, alignment check between the upstream and downstream telescope arms

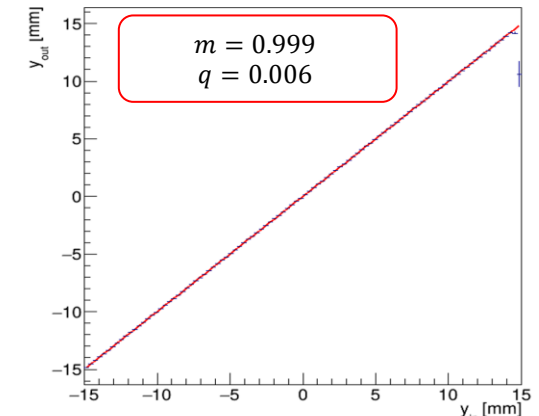
To ensure that the upstream and downstream detectors are aligned, we check the deviation between the initial and final positions of particles that definitely did not hit the crystal, ensuring that is zero.



x_{out} vs x_{in} (air only)



y_{out} vs y_{in} (air only)



We can also check the alignment by doing a linear fit between the upstream and downstream profiles.

ANALYSIS: CRYSTAL EDGES

PRELIMINARY ANALYSIS

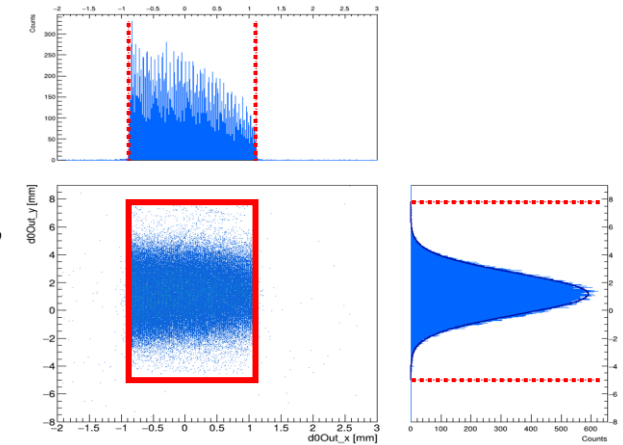
2

CRYSTAL EDGES

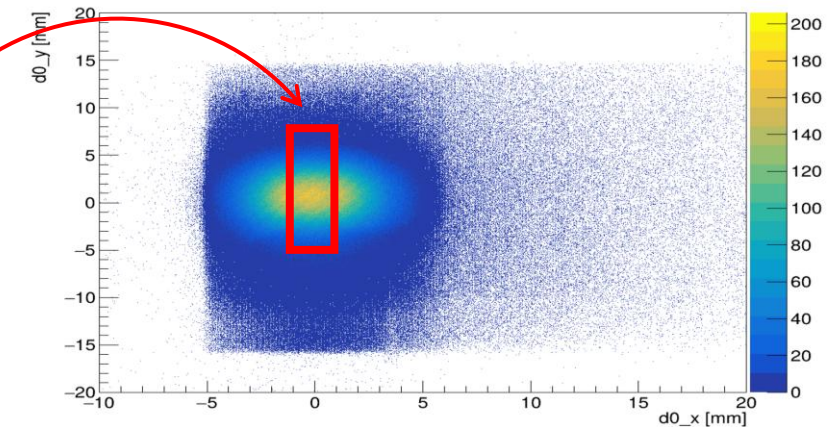
Reconstruct the (x,y)
spatial window
illuminated by the
crystal
(2 independent
methods)

METHOD 1: CHanneled PARTICLES

- Isolate already channeled particles (only uses ~ 20% of data)
- Their impact position (x, y) traces the crystal footprint, matching its expected dimension
- Use of a sliding window method, which maximize the number of particles inside
- To project the downstream edges back upstream a shift is needed (due to the lateral geometric drift along the x-axis)



Selected window
2 mm × 12.8 mm



METHOD 2: MULTIPLE COULOMB SCATTERING

ANALYSIS: CRYSTAL EDGES

PRELIMINARY ANALYSIS

2

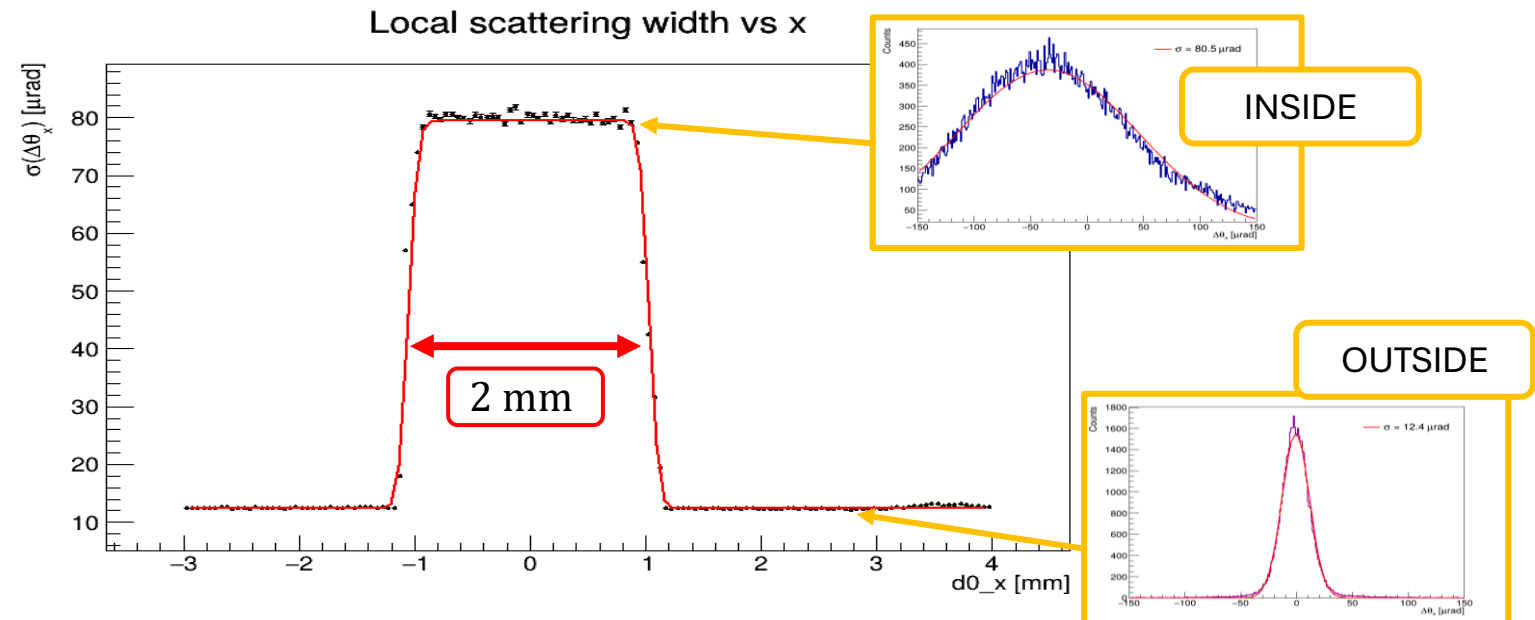
CRYSTAL EDGES

Reconstruct the (x,y)
spatial window
illuminated by the
crystal
(2 independent
methods)

METHOD 1: CHanneled PARTICLES

METHOD 2: MULTIPLE COULOMB SCATTERING

- Particles that physically cross the crystal picks up extra Coulomb scattering, the ones outside the crystal do not
- The width σ of the outgoing angular distribution is measured
- A sharp transition marks the edges



ANALYSIS: MAPPING THE TORSION

PRELIMINARY ANALYSIS

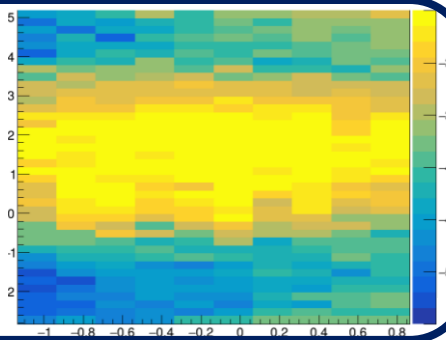
3

TORSION MAP

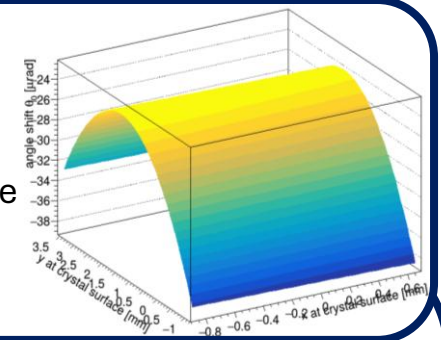
2D checkerboard grid analysis of the entrance face:
surface fit to obtain
 τ_x, τ_y

Beam spot divided
into a 0.1 mm (x,y)
grid

Angular scan
for each cell
→ local θ_0
that maximize
efficiency



Surface fit
(parabolic)
to extract the
global τ_x, τ_y



RESULTS:

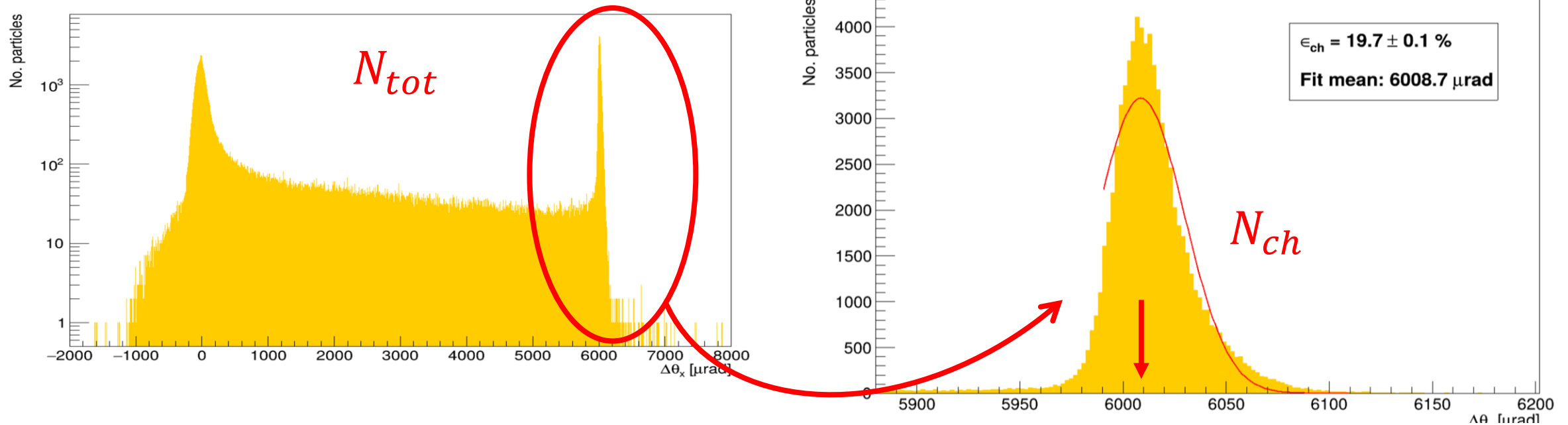
Torsion along the y -axis: $\tau_y = 5.98(3) \mu\text{rad/mm}$
Negligible torsion along the x -axis: $\tau_x = 0.1(1) \mu\text{rad/mm}$

CHANNELING EFFICIENCY

ANALYSIS

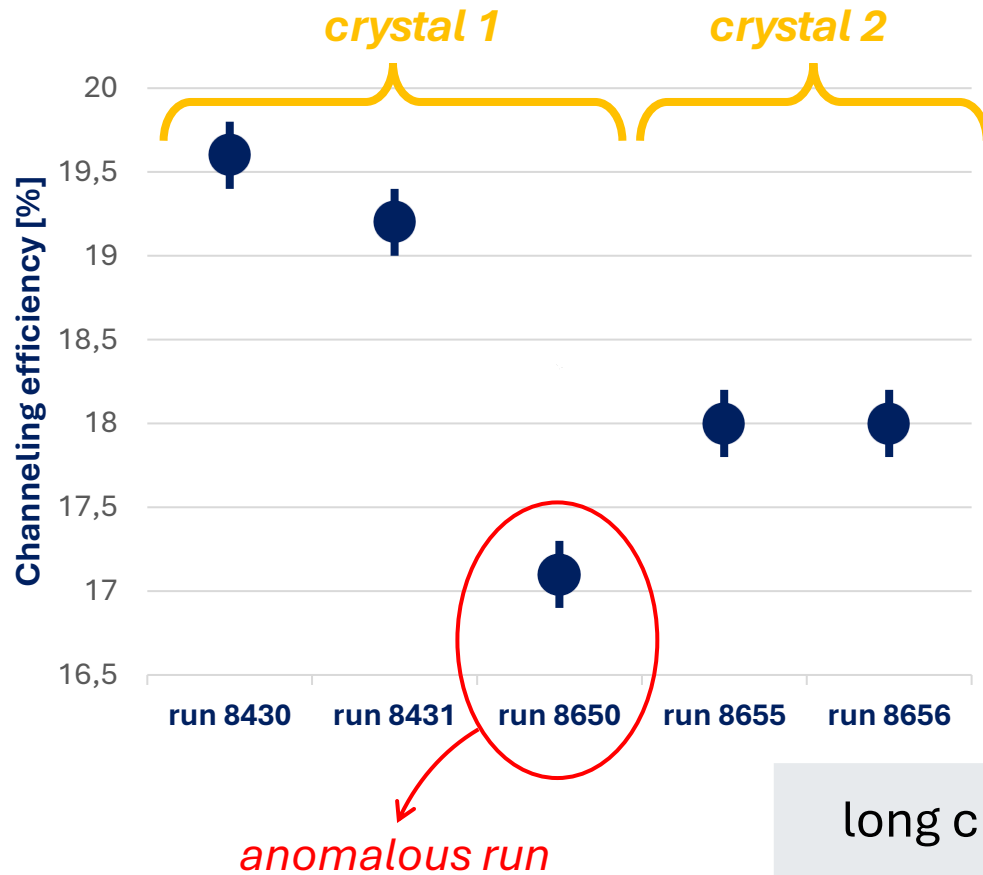
CHANNELING EFFICIENCY

$$\epsilon_{ch} = \frac{N_{ch}}{N_{tot}}$$



CHANNELING EFFICIENCY: RESULTS

ANALYSIS



RUN 8430 (REFERENCE)

$(19.6 \pm 0.1 [\text{stat}] \pm 0.2 [\text{syst}])\%$

	ϵ_{ch}	τ_y
long crystal 1	$(19.4 \pm 0.1 \pm 0.2)\%$	$(5.98 \pm 0.03)\mu\text{rad/mm}$
long crystal 2	$(18.0 \pm 0.1 \pm 0.2)\%$	$(3.77 \pm 0.01)\mu\text{rad/mm}$

CHANNELING EFFICIENCY: RESULTS

ANALYSIS

Long crystal 1

$(19.4 \pm 0.1[\text{stat}] \pm 0.2[\text{syst}])\%$

Long crystal 2

$(18.0 \pm 0.1[\text{stat}] \pm 0.2[\text{syst}])\%$

ERROR BUDGET

Statistical error:

- Binomial error
- Fit errors

0.1%

Systematic error:

- Fit model choice
- Binning choices

0.2%

CHANNELING EFFICIENCY: RESULTS

ANALYSIS

long crystal 1	$(19.4 \pm 0.1 \pm 0.2)\%$	$(5.98 \pm 0.03)\mu\text{rad/mm}$
long crystal 2	$(18.0 \pm 0.1 \pm 0.2)\%$	$(3.77 \pm 0.01)\mu\text{rad/mm}$
	ϵ_{ch}	τ_y



On article *Bandiera et al.* (2025):

It was found and efficiency of $\epsilon_{ch} = (15.8 \pm 0.1)\%$ and a torsion of $\tau_y = (24.4 \pm 0.2)\mu\text{rad/mm}$ for the TCCP long crystal.

Instead our two crystals can reach a maximum efficiency of 23 – 25% in some areas.

WHAT'S NEXT: SIMULATION

NEXT STEPS



Set up a Geant4 simulation of the long crystals characterized in this analysis, reproducing the same geometry as the H8 setup.



Geant4/BDSIM:

- Xtrack along the beamline
- Geant4 only for elements where there is interaction with matter (collimator, crystals,...)



The actual data serve as a **benchmark**.



The simulation reproduces the “ideal” channeling physics, without torsion or other imperfections → simulated efficiency is expected to be **higher**.



The gap between the simulation and the data serves as a quantitative measure of the impact of torsion and other non-idealities.

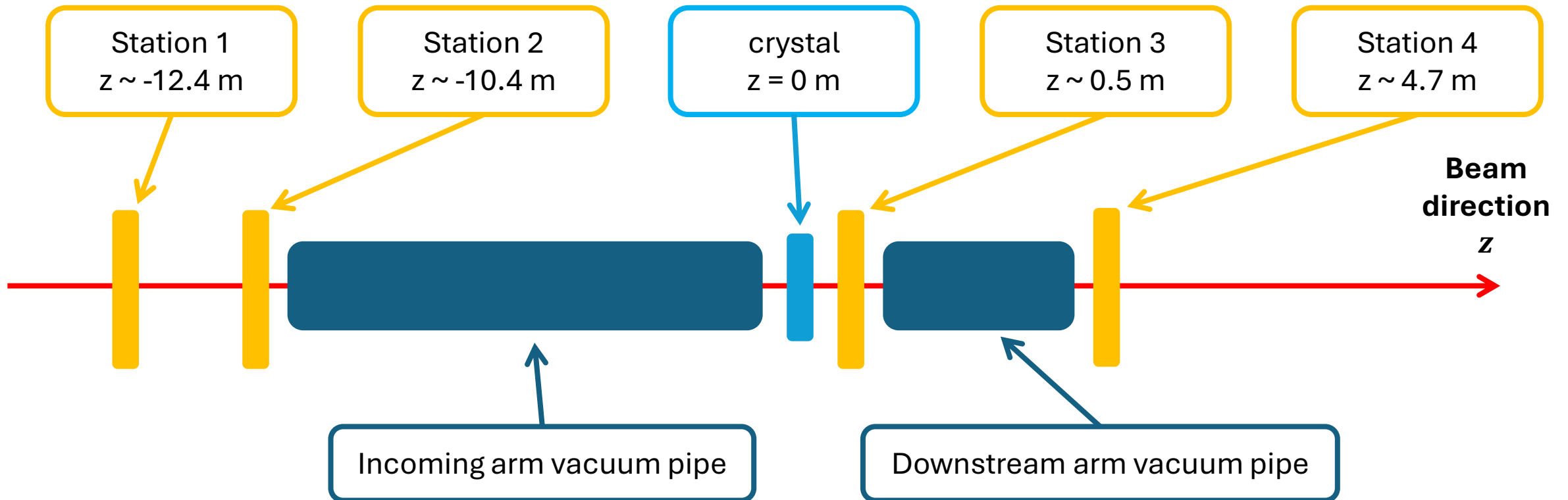
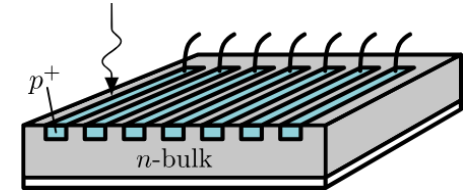
BACKUP SLIDES

SCHEMATIC LAYOUT

BACKUP SLIDE

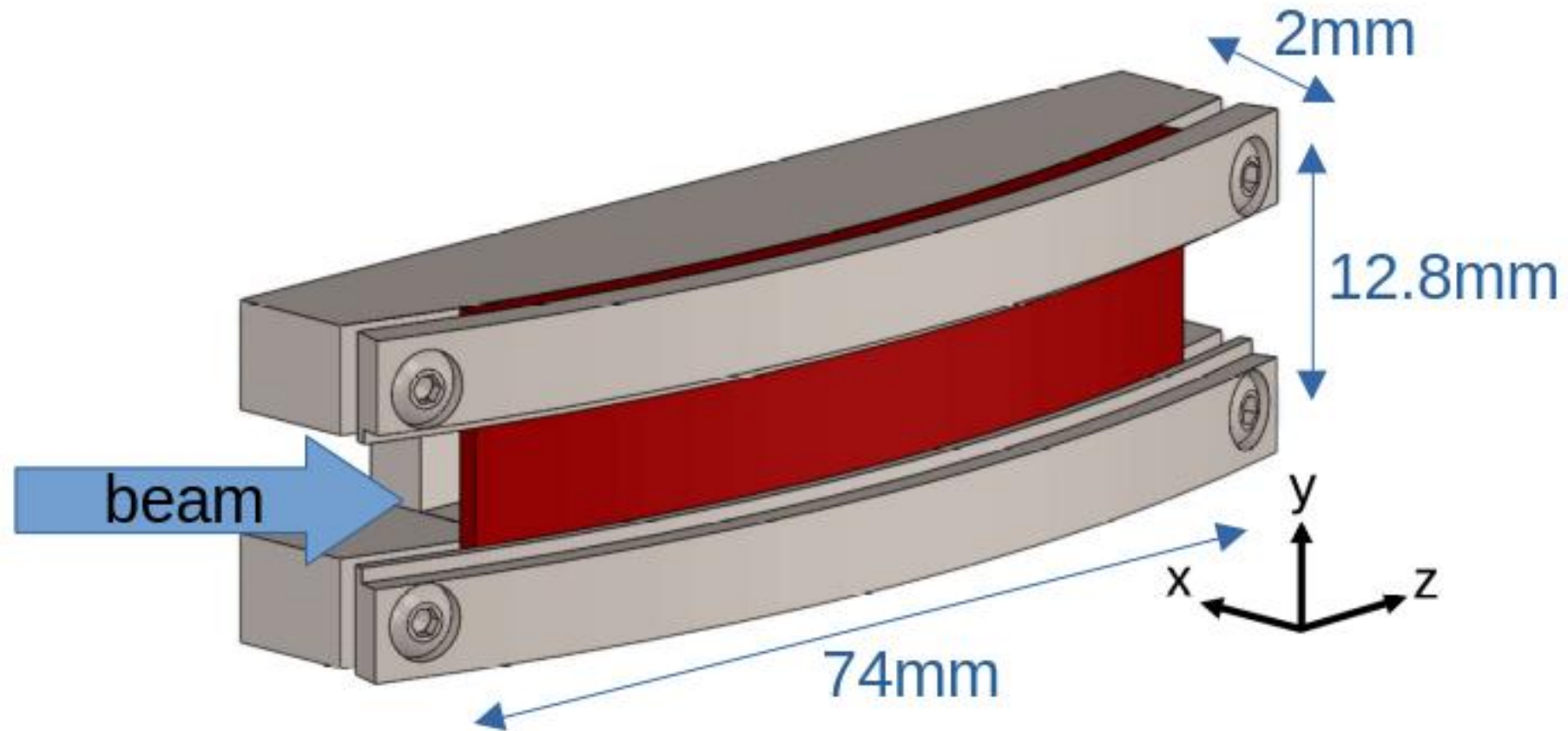
Beam: H8 secondary beamline off the SPS, driven by 400 GeV protons
Spectrometer selection 180 GeV hadrons (30% π^+ and 70% p)

Trackers:
2 silicon microstrip sensors
639 strips, 98 mm length
with a ~ 7 μm resolution



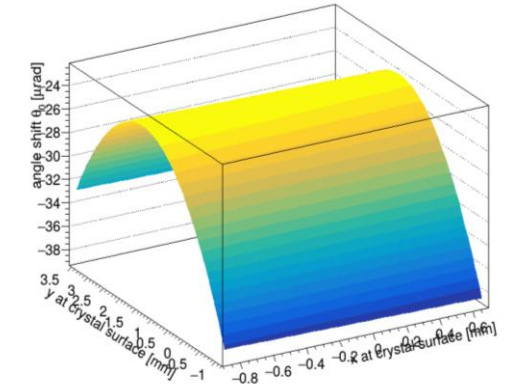
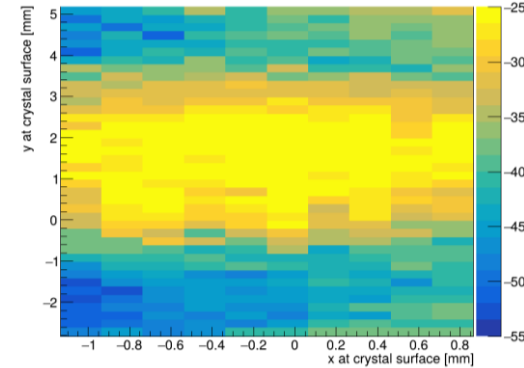
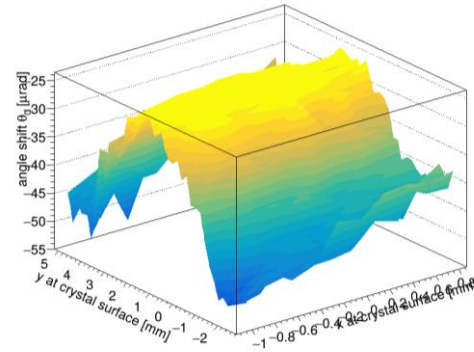
SCHEMATIC LAYOUT

BACKUP SLIDE



TORSION FIT

BACKUP SLIDE



The 2d map is fitted through a parabolic function:

$$\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 y^2$$

Now we correct the effect of torsion using the parameter $\theta_0(x, y)$ found for each bin:

instead of the Lindhard cut $|\theta_{in,x}| \leq \frac{1}{2} \theta_L$ we will use

$$\left| \theta_x^{in} - \left(\theta_x^{off} + \tau_y \cdot d_y + \tau_x \cdot d_x \right) \right| \leq \frac{1}{2} \theta_L$$

ANOMALOUS RUN 8650

BACKUP SLIDE

Due to an issue, the run was taken at 150 GeV instead of 180 GeV.

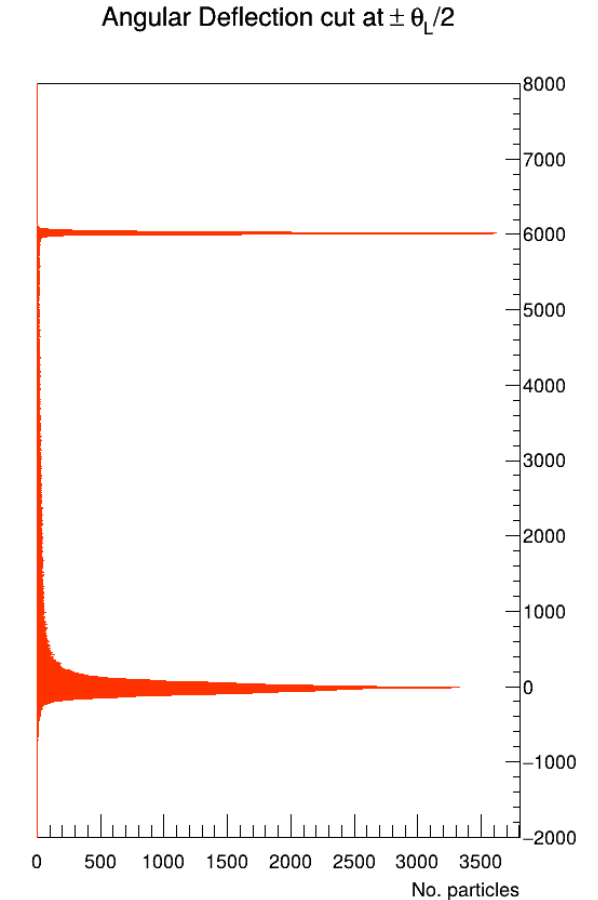
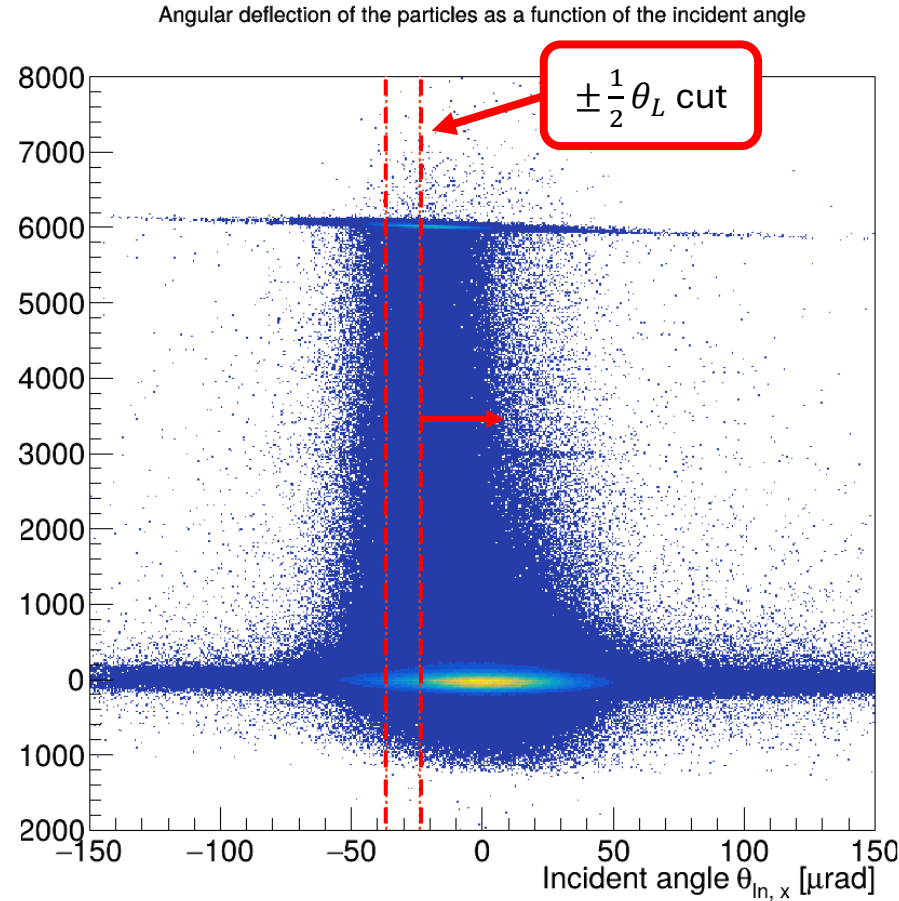
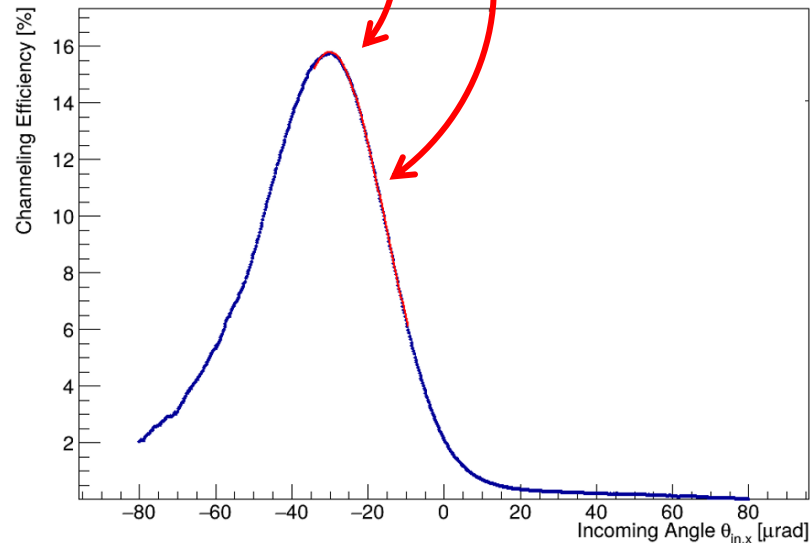
- Wider channeling peak
 - More multiple scattering ($\propto 1/p$)
- Lower efficiency
 - De-channeling more likely at lower energies
- Larger critical Lindhard angle

ANGLE SELECTION

BACKUP SLIDE

For each square bin:

- variation of θ_0
- select particles within $\leq \frac{1}{2} \theta_L$
- compute ϵ_{ch}
- select θ_0 that maximizes ϵ_{ch} (through gaussian fit)



EFFICIENCY CURVE

BACKUP SLIDE

Crystal efficiency as a function of the incoming angle, after application of the torsion correction. The value of the peak is the efficiency of the crystal when it is perfectly aligned to the beam.

